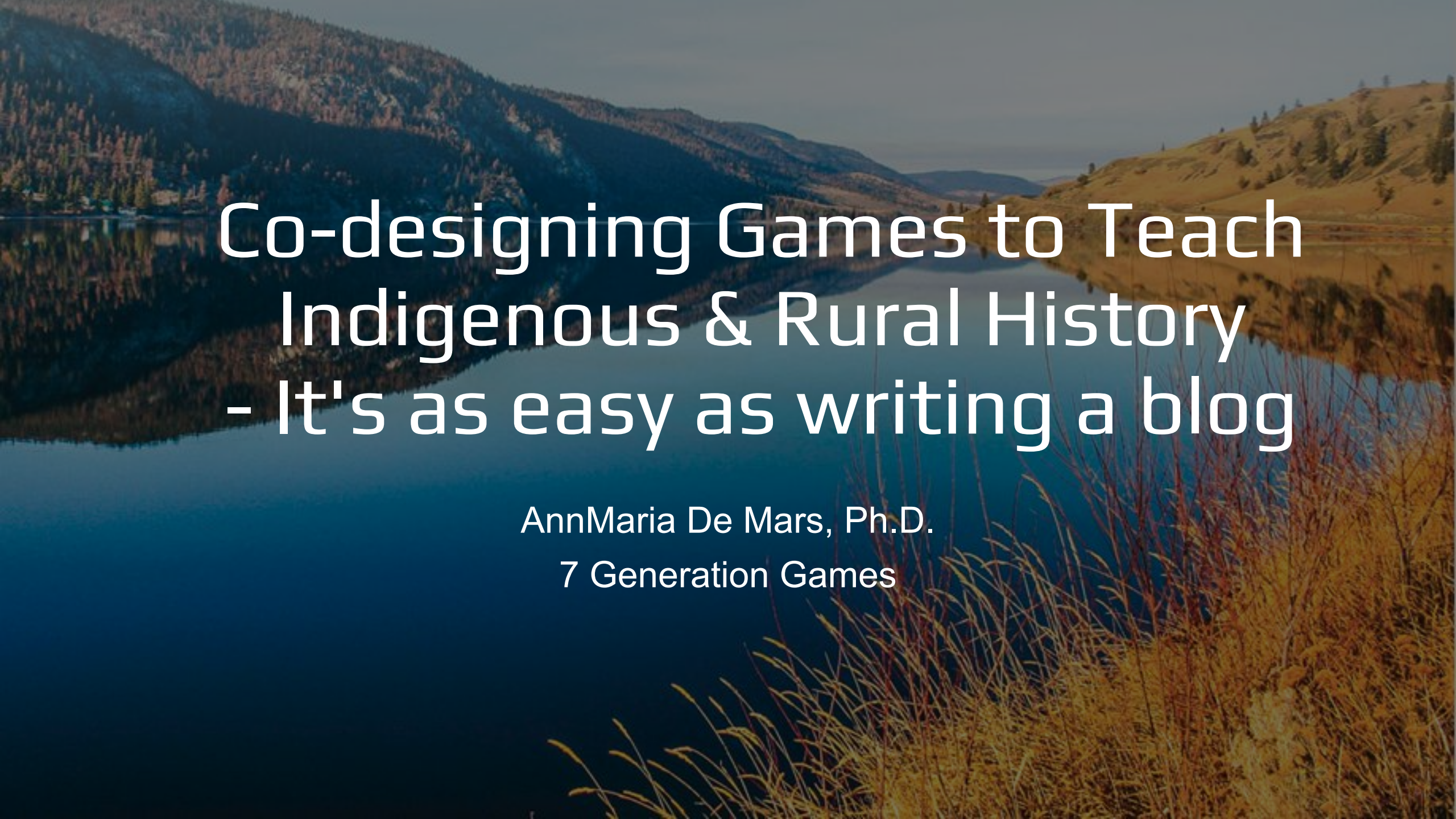




Co-designing Games to Teach Indigenous & Rural History - It's as easy as writing a blog

AnnMaria De Mars, Ph.D.

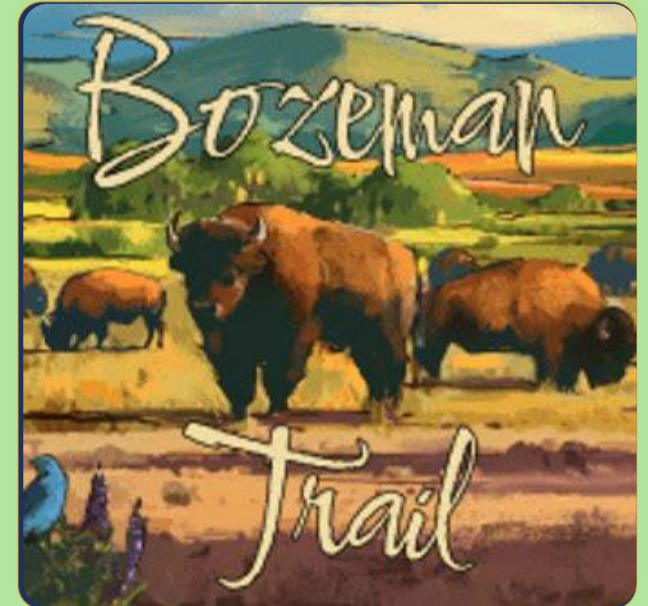
7 Generation Games



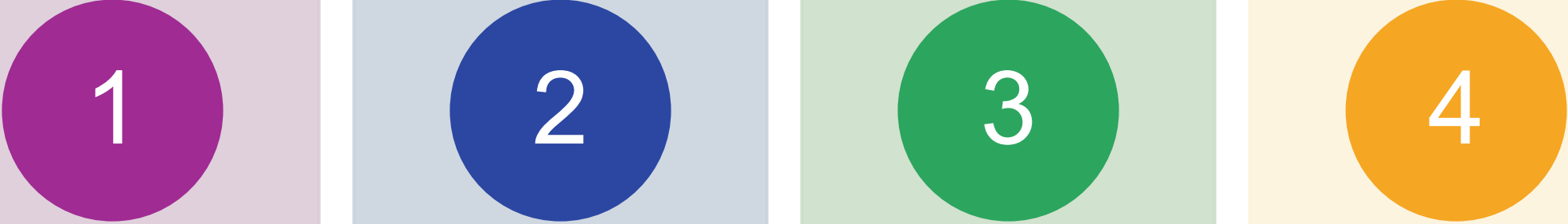
ANNMARIA DE MARS, PH.D.

- 7 GENERATION GAMES CO-FOUNDER
- 7 GEN BLOCKS DEVELOPER
- MATH TEACHER & PROFESSOR

annmaria@7generationgames.com



My goals



1

Show examples of teaching Indigenous history using games (Ojibwe, Lakota, Dakota, Navajo, Maya)

2

Walk you through creating a game design to make your own games

3

Introduce a resource you or your students can use to make games

4

Participate in user interface/user experience (UI/UX) design

A wide-angle landscape photograph of a calm lake in the Upper Nicola Valley. The lake's surface is a deep, clear blue, acting as a perfect mirror for the surrounding environment. On the left, a steep, forested mountain rises from the water's edge, its dark green and brown tones reflected in the water. On the right, a gentler, golden-brown hill slopes down towards the lake, also reflected. In the foreground, the bottom right corner is filled with tall, dry, golden-brown grasses and reeds, some of which are slightly out of focus. The sky above is a pale, hazy blue with soft, wispy clouds. The overall atmosphere is peaceful and serene, capturing a beautiful natural scene.

Our story began in the Upper Nicola Valley

- A problem almost EVERY teacher has, no matter how gifted, is students who just do not do their work or pay attention in class.
- One solution is to create math games that include math in context, like how many of this herb to collect if 24 people are sick
- Games that are "classroom-ready" and integrated lesson plans ready to use.



Growing Math

Classroom-ready learning resources for Grades 3-8

[LESSONS](#)[DATA](#)[GAMES](#)[RESOURCE LIBRARY](#)[FAQ](#)[CONTACT](#)[TRAINING](#)

Developed with



growingmath.org

7generationgames.com

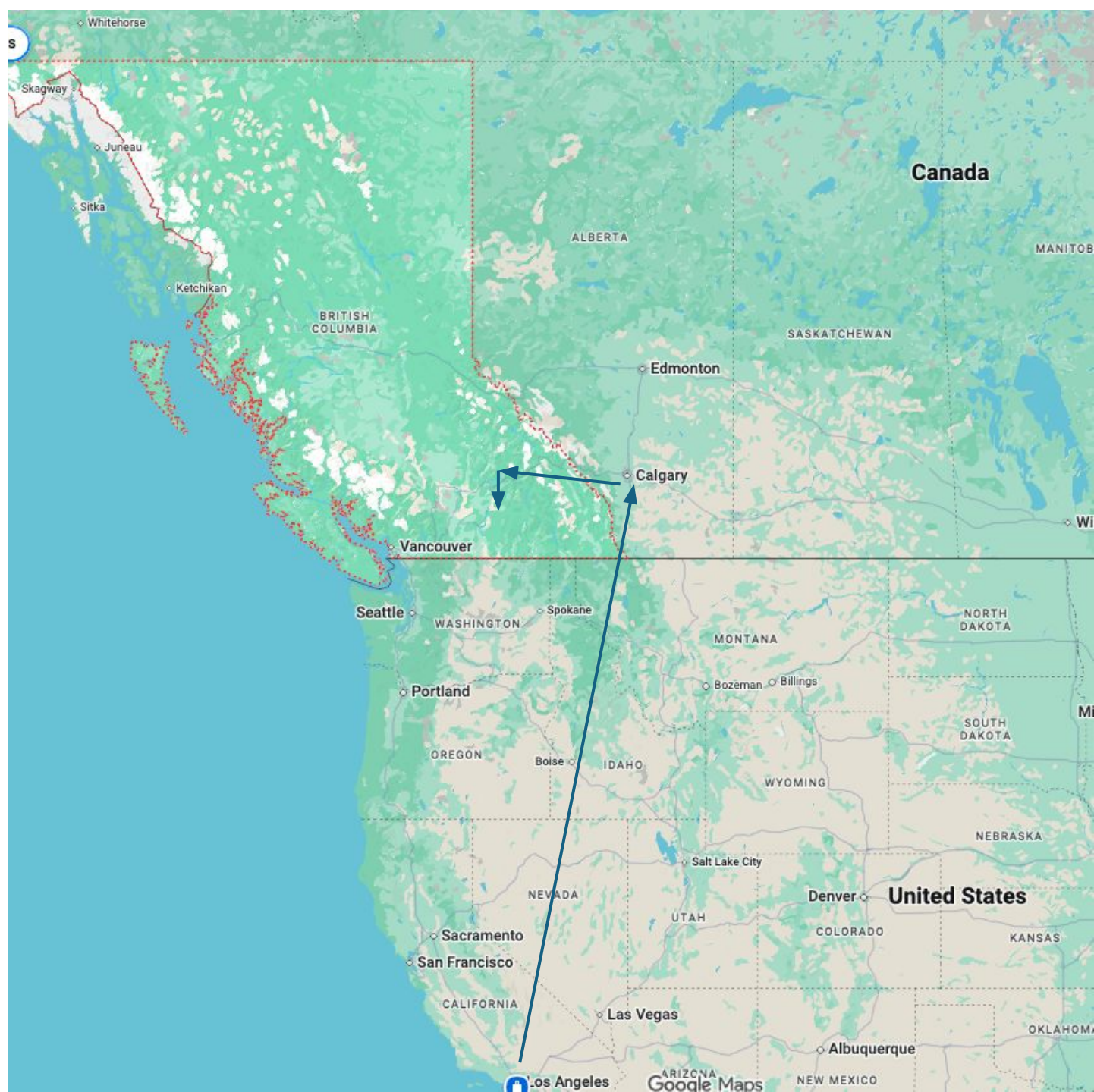
MAKING CAMP

LAKOTA

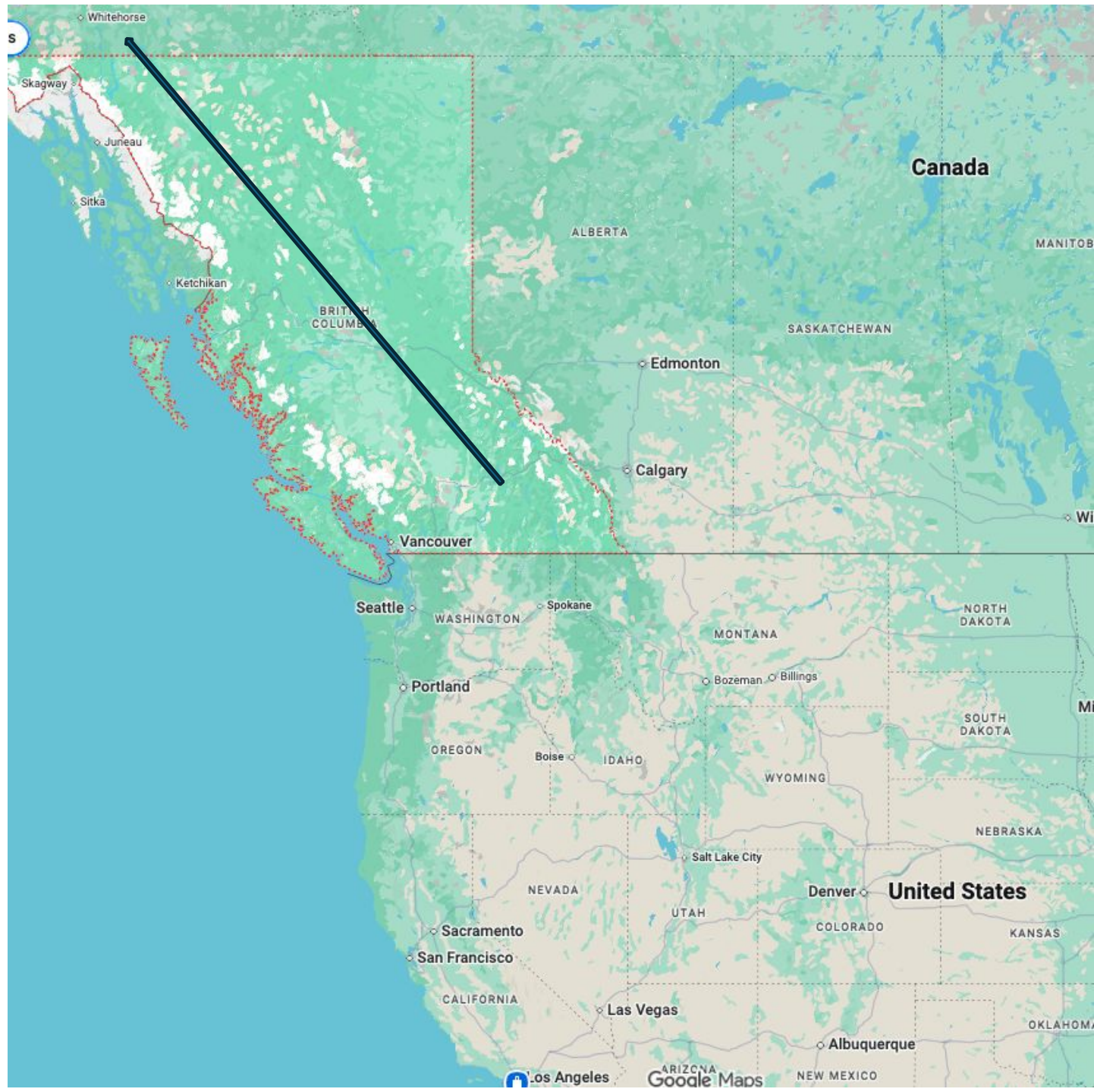


7th Generation Games

"Can you stay?
Some people from
way up north want
to talk with you?"



7 Gen Blocks came about because I could not help the teachers way up north revitalize their language.







Children learn while playing*

“If making an educational game was as easy as writing a blog, educators could create games that reflected their community and language.”

* Adults do, too.



The solution: games that move the numbers

30%

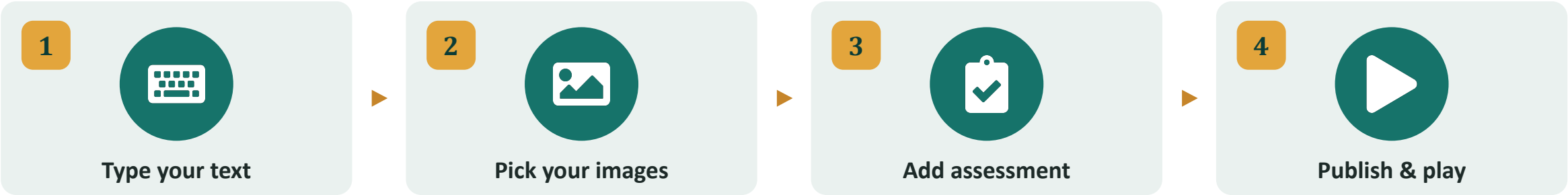
math score gain in 10 weeks

3×

the gain of the control group

2

languages: English + Spanish/ Indigenous languages



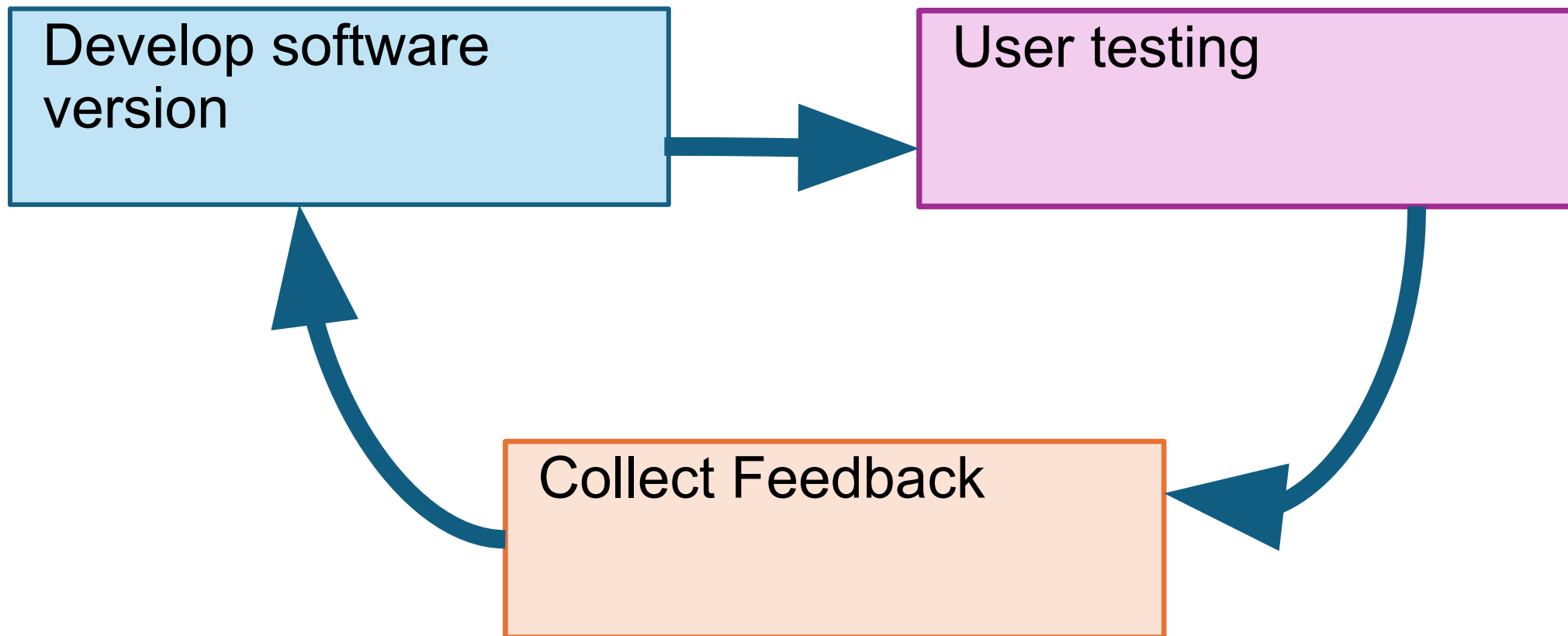
*Live no-code
builder*

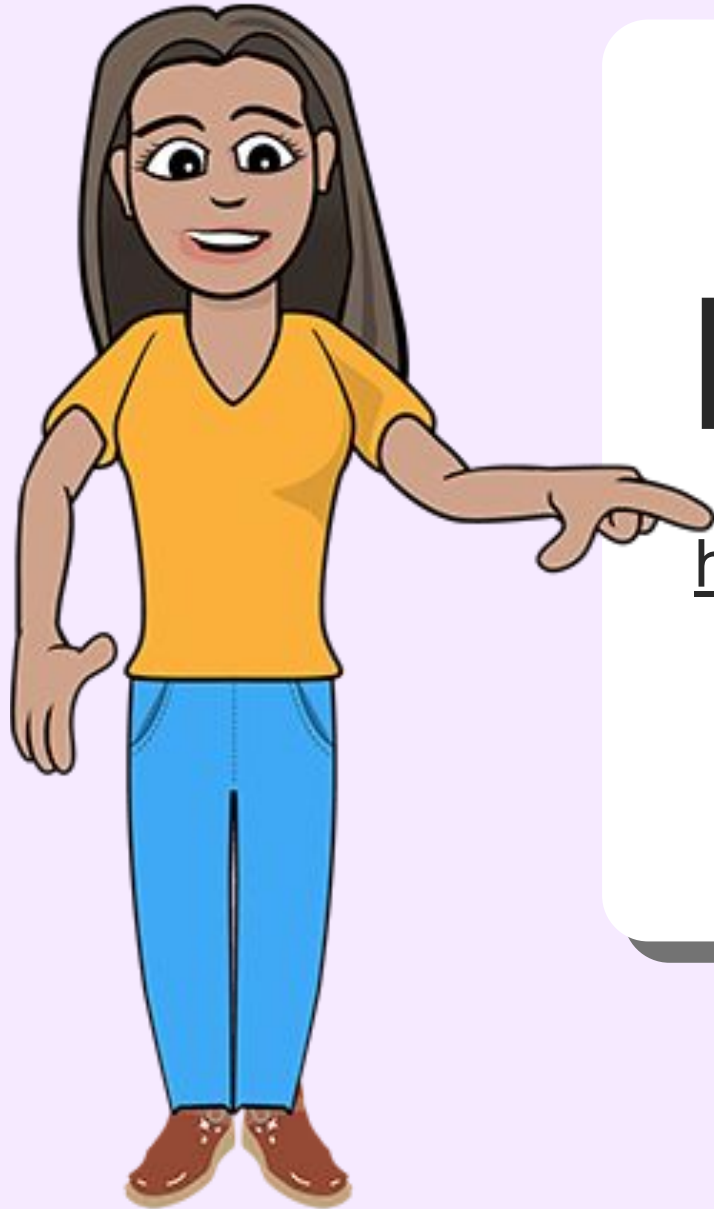


No logins. No installs.

Runs on low-cost devices and low-bandwidth connections.

Our process





Here is the link!

<https://game-builder-7gen.com/login>

7 GEN BLOCKS GAME BUILDER

Snap together virtual blocks to make educational games without code.

What do you want to do?

- ☐ Create a New Game
- ☐ Edit an Existing Game



7 GEN BLOCKS GAME BUILDER

Snap together virtual blocks to make educational games without code.

1. Name your game

Game Name:

SUBMIT

NOTE: Your game will be saved as this name.html, all lower case and with no spaces, e.g. rocks_from_mars.html

7 GEN BLOCKS GAME BUILDER

Snap together virtual blocks to make educational games without code.

Your game name is rocks_from_mars

The URL for your game is null/rocks_from_mars

2. Pick a Theme



● Old-fashioned: Uses Copperplate font, grey and black icons



● Cartoon (best for younger players): larger font, two lines per screen in visual novel



● Mid (best for middle school)

7 GEN BLOCKS GAME BUILDER

Snap together virtual blocks to make educational games without code.

Your game name is molokai

The URL for your game is <https://game-builder-7gen.com/static/users/annmaria/molokai>

Choose Your Block



- Splash Screen Block
- Drag into Categories Assessment Block
- Level Up Screen Block
- Matching Terms Assessment Block
- Moving Targets Game Block
- Runner Game Block
- Visual Novel Block
- End Game Block

Your theme is - Mid

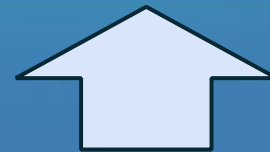
7 GEN BLOCKS GAME BUILDER

Splash Block

This is the screen your players will see when they start a game. It will have a background image and a title, which can be text or an image.

What is the name of your splash block? We recommend something basic like "splash" or "start".

SUBMIT



7 GEN BLOCKS GAME BUILDER

Splash Block

This is the screen your players will see when they start a game. It will have a background image and a title, which can be text or an image.

Your block name is start. Your block url is /start.html

Use the Asset Manager to choose the background image. You can choose an existing image or upload a new one.

Asset Manager



Filter by tags:

Select Tag from List

Clear

All Tags

Any Tag



lake



buffalo_on_hill



splash



gloria



splash.jpg



scene1



title_pyramid_inside.jpg



scene2.jpg



scene3.jpg



scene3.jpg



scene4.jpg



scene5.png

Showing 1 to 12 of 17 results

Upload Asset

<

1

2

>

Click to select from images provided or upload your own images.

7 GEN BLOCKS GAME BUILDER

Splash Block

This is the screen your players will see when they start a game. It will have a background image and a title, which can be text or an image.

Do you want to superimpose a text title over the image? (If the image already contains a title, don't select this. Just click SUBMIT)

☐ Use title text

What is the title?

What is the title text color?

SUBMIT



7 GEN BLOCKS GAME BUILDER

Splash Block

This is the screen your players will see when they start a game. It will have a background image and a title, which can be text or an image.

Do you want to superimpose a text title over the image? (If the image already contains a title, don't select this. Just click SUBMIT)

☐ Use title text

What is the title?

What is the title text color?

SUBMIT

If your image already includes the text for the title, do NOT click the box next to title text. If you want to add text over the image, click the Use title text box, then type the text you want for the title. If you want a color other than black for the title text, click the box with the color and select the color you want. Click submit to save these changes and go on to the next screen.

OK



Cancel

?



7 GEN BLOCKS GAME BUILDER

Splash Block

This is the screen your players will see when they start a game. It will have a background image and a title, which can be text or an image.

What is the name for the next block? This should be something like: runner or find_the_cheese. Do not include the directory or .html

SUBMIT

It doesn't matter if this block doesn't exist yet. Also, you can always come back here later and change the name if you change your mind.



Once you've learned one block, you've
learned most of what you need to know

7 GEN BLOCKS GAME BUILDER

Moving Targets

A mini-game in which players click on 'good' items to catch them, like salmon. There may also be 'bad items' to avoid, like old shoes.



Creating Moving Targets block for game: test_game_title

What is the name of your block? We recommend something easy for you to remember and descriptive, like "Catch Salmon".

SUBMIT



7 GEN BLOCKS GAME BUILDER

Moving Targets

A mini-game in which players click on 'good' items to catch them, like salmon. There may also be 'bad items' to avoid, like old shoes.

Creating Moving Targets block for game: test_game_title

Your block name is catch_bugs. Your block url is test_game_url/catch_bugs.html

The next five questions are about the overall game setup. After that, you'll give information on the targets.

What are the instructions for the game? For example, "Click on 8 salmon to win. Leave the other fish".

SUBMIT



7 GEN BLOCKS GAME BUILDER

Drag Items into Categories

An assessment block. Drag the items at the bottom to the appropriate category at the top.

Creating Drag into Categories block for game: diving_decimals

What is the name of your block? We recommend something easy for you to remember and descriptive, like "drag_tribal_perspective".

SUBMIT



?

Match the views with the party in this situation whose perspective it represents.



N. Arapaho



E. Shoshone



U.S. Government

Consistently revised treaties, often unilaterally, when they didn't want to follow them.

Were being asked to share their hard-won Reservation with an adversary.

Wanted to keep their people safe on lands promised through the 1868 Fort Bridger Treaty.

Had to rely on an adversary to share their lands with them.

Correct: 1
Incorrect: 0

7 GEN BLOCKS GAME BUILDER

Snap together virtual blocks to make educational games without code.

What do you want to do?

- ☐ Create a New Game
- ☐ Edit an Existing Game



7 GEN BLOCKS GAME BUILDER

Game: diving_decimals

Delete This Game

You can preview the game you are building at: https://game-builder-7gen.com/static/users/annmaria/diving_decimals/index.html

Edit an Existing Block

This is a list of blocks in your game. Click on a row to edit or delete a block.

Component Name	Block ID	Component ID
splash	Splash Page	232
moving_fishes	Moving Targets	234
decimal_drag	Drag into Categories	239

Create a New Block:

Choose a block to create from the dropdown list below.

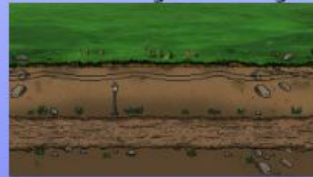


Editing is easy

Block Type: Runner
Component Name: runner
Component URL: <https://game-builder-7gen.com/static/users/annmaria/msm/runner/runner.html>

Instructions	Try to get your car to the wharf so you can get to work. USE THE SPACE BAR TO JUMP.
Collision Counter Label	Rock Collisions
Win Game Message	Your car, and you, survived the road. Now, let's build that wharf.
Collision Limit	5
Message for too many collisions	Sorry, you have a flat. You hit too many rocks.
End game obstacle count	8
Ground Level	150px
Runner Height	50px
Completion Block	math_facts

Background Image
(click on image to change)



Runner Image
(click on image to change)



Goal Image
(click on image to change)



Obstacle Image
(click on image to change)



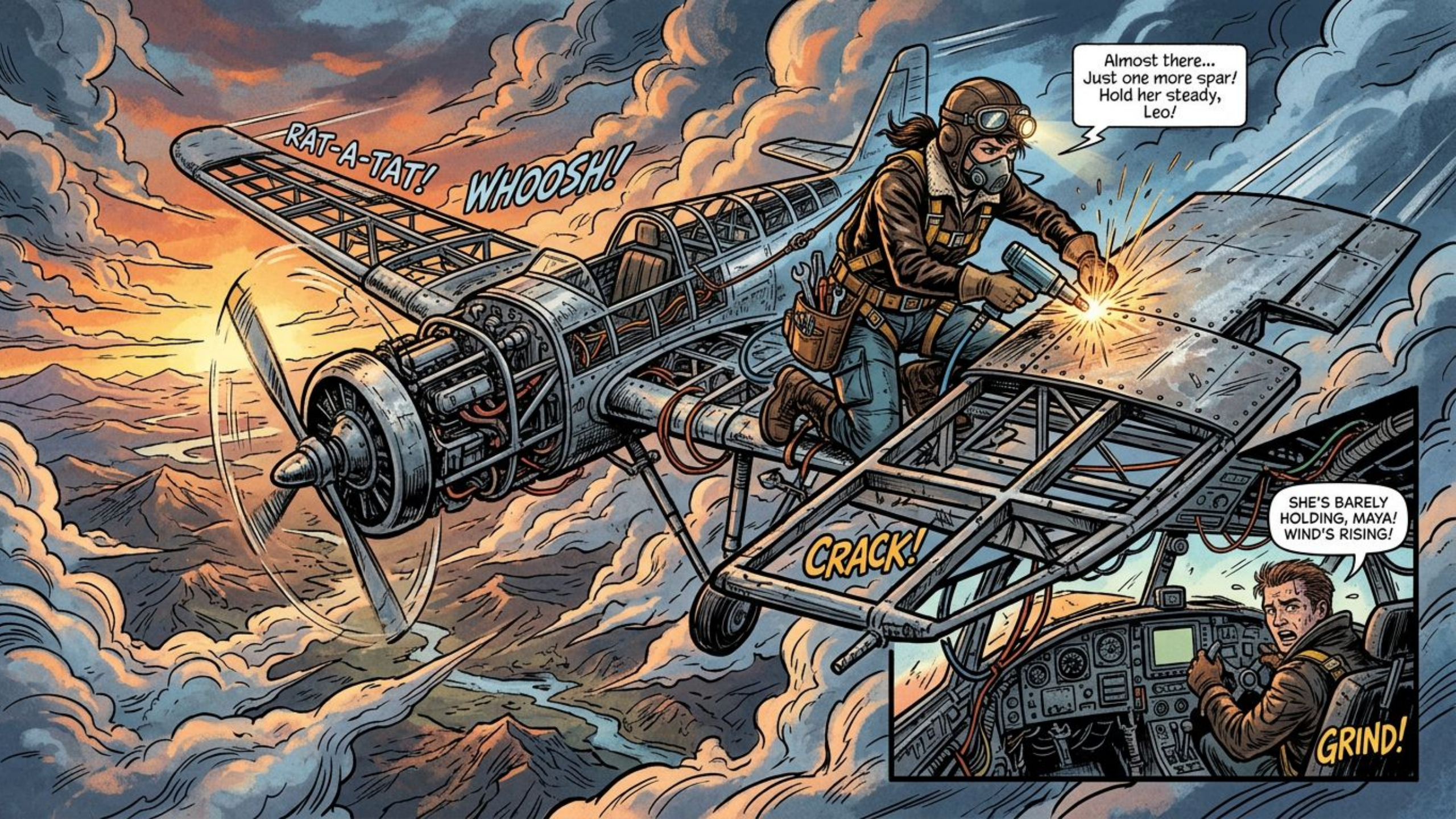
Submit Options Edits

Create Another Block

Create/Edit Another Game

Log Out





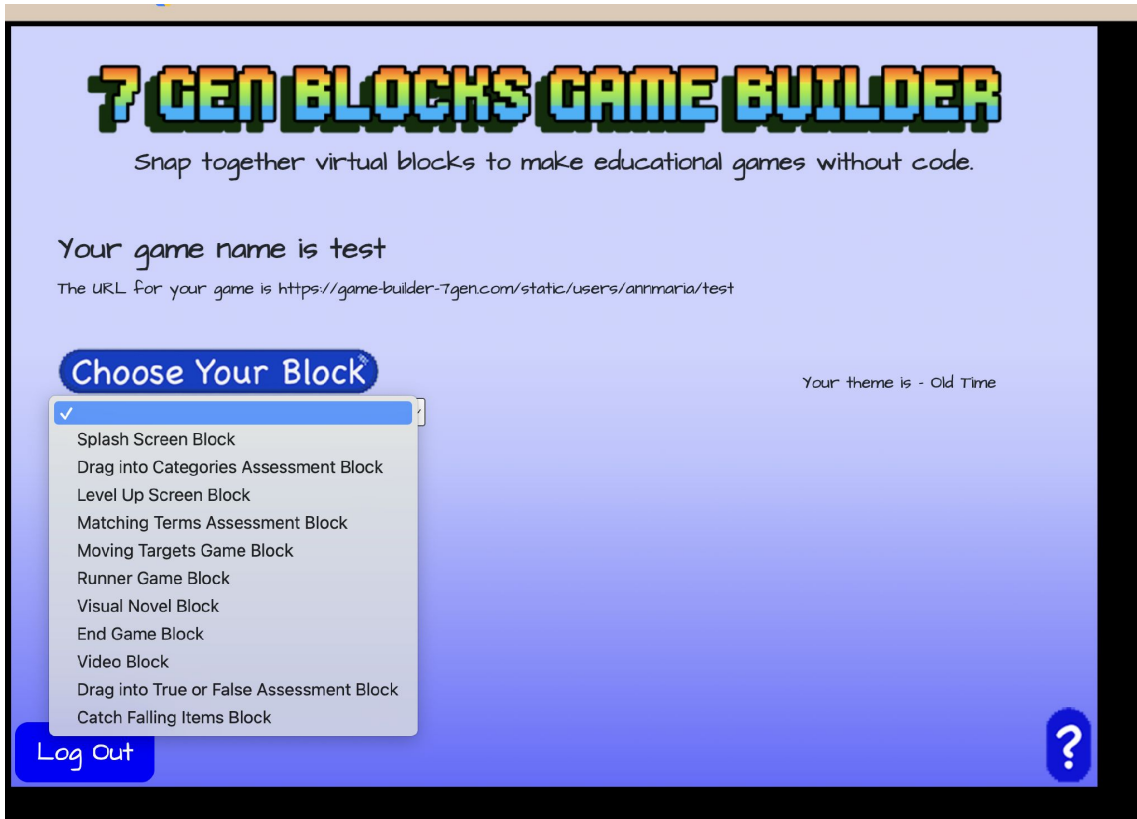
RAT-A-TAT! WHOOSH!

Almost there...
Just one more spar!
Hold her steady,
Leo!

CRACK!

SHE'S BARELY
HOLDING, MAYA!
WIND'S RISING!

GRIND!



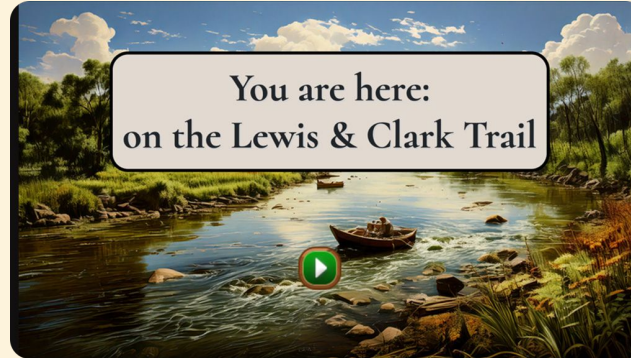
Changes in response to user testing can be minor

Splash screen = your title page = start page

The first thing players see, every single time they start your game. Make it count.



A hook



A world



A welcome



7 GEN BLOCKS GAME BUILDER

Snap together virtual blocks to make educational games without code.

What do you want to do?

- ☐ Create a New Game
- ☐ Edit an Existing Game



The solution .. But first ...

What you need to make a game



Create a game design with AI



Create images with AI



Your brain + AI: the design doc combo



Gemini

Copilot

ChatGPT



Good design docs come straight from your brain, or your brain plus AI. Don't just paste in whatever AI gives you.

It's the puzzle that matters — AI just helps with the pieces

Optional: start with a prompt

CONTEXT

I need to create an educational game that runs on Chromebooks for middle school students, using a game builder platform that has blocks of code for each game component.

INSTRUCTIONS

Except for the splash screen and end game block, it is acceptable to use a block more than once.

PROMPT

Write an educational game design that has the history of the Miwoks as the game narrative and teaches the following math standards:

CCSS.Math.Content.6.NS.C.7 — Understand ordering and absolute value of rational numbers. Use only the blocks found here. Begin with a splash screen and end with an end game block. Use these links provided as reference material on Miwok history

<https://en.wikipedia.org/wiki/Miwok>

<https://www.coastmiwokofmarin.org/our-history.html>

<https://angelisland.org/history/miwok-history/>

THE BLOCKS

- Splash screen
- Drag into Categories Assessment
- Level Up
- Make a Visual Novel
- Matching Terms
- Moving Targets Game
- Runner Game
- End Game Block

Copied from: 7genblocks.com/blocks-edu/build-with-blocks-edu

Are we making a game yet?

One more step ... get your artwork together.



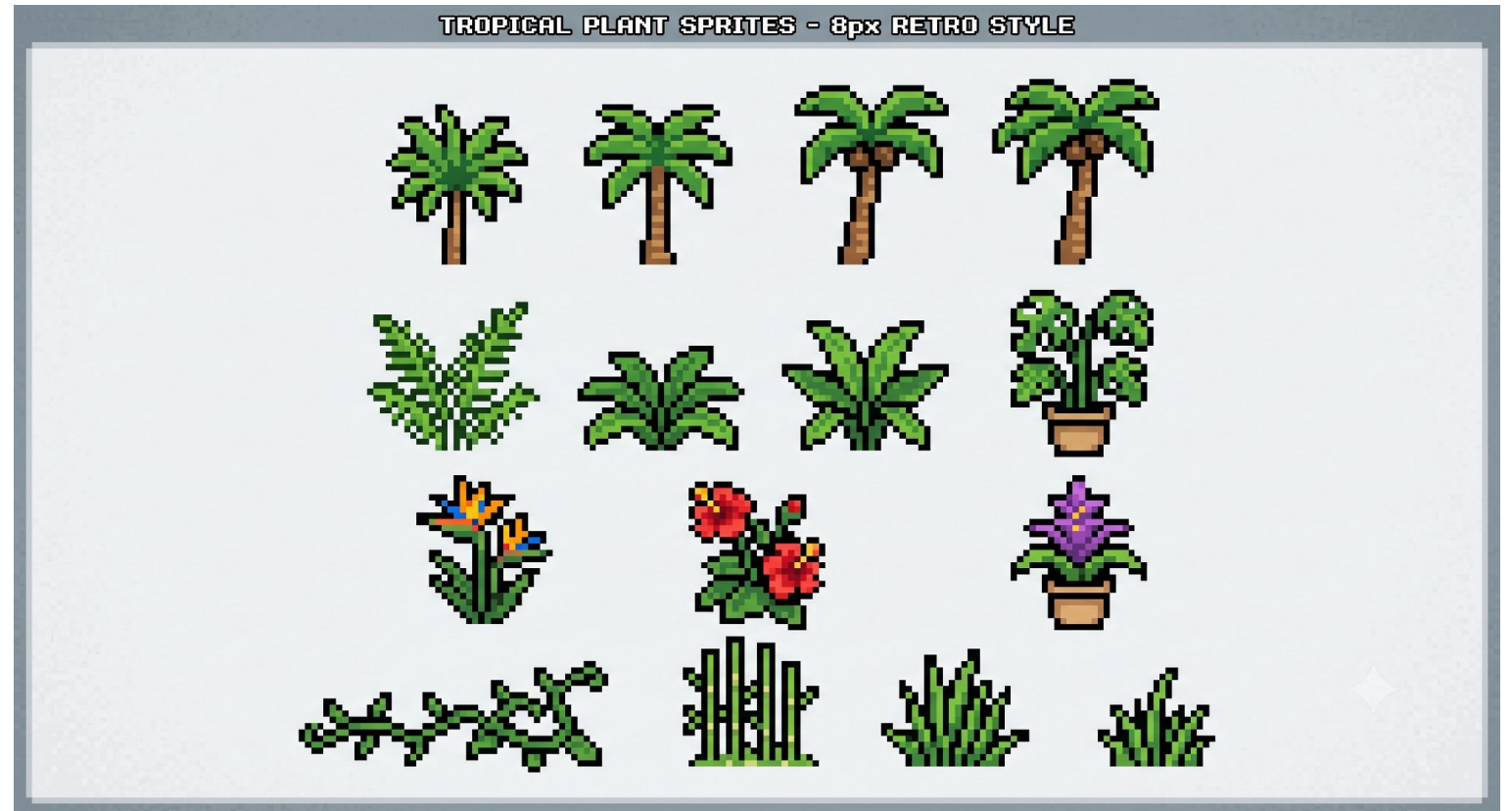


Four places to get artwork

- Create it yourself if you have talent (I don't)
- Download it from public domain sources (Library of Congress is great)
- Have your students create artwork
- Create it with AI (Adobe Firefly, Gemini, Copilot)
- Consider using photos as reference

PROMPT: Tropical plants to be used in a video game in 8px retro style

Pro Tip:
Save things



Images Audio Video

Filter by tags: Select Tag from List Clear All Tags Any Tag



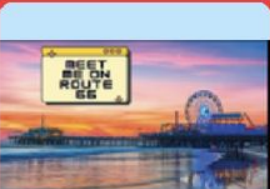
campfire page one



explosion



girl_boy_pier1



splash66



firework



angie



beach_background



intro_sm_woman



jellyfishes



old santa monica



mountains_sunset



wharf_numbers

Showing 1 to 12 of 80 results

< 1 2 3 4 5 6 7 >

Upload Asset Cancel

Making it easier and easier



jellyfishes 

PNG File
2.8M
1376w X 768h

Tags 

Select Delete

Aztec Pack

Unlike some of our packs, none of the Aztec art is AI-generated and it is one of our larger packs. This is all created by our artists, Justin Flores and Stephen Gladue. There are 17 background images which include scenes, designs and city backgrounds. There are also ten larger characters representing different levels of Aztec society, from the lowest classes to the soldiers and god-king. This pack also includes four smaller characters that could be used in a runner game. [Download the Aztec pack.](#)



Beach Pack

This pack, created by Diana Sanchez, using Adobe Express, includes 8 poses of human characters, 2 animal characters, 13 background scenes for visual novels and 3 backgrounds for runner games. [Download the beach pack.](#)



Games Teaching Indigenous & Rural History with Primary Sources

Thanks to the Library of Congress Teaching with Primary Sources grant from the western region, we were able to meet with cohorts of teachers to co-design games. The first seven are completed. Four of these completed games – Follow the Bubbles, The Greasy Door, Healing Journeys and Plant Knowledge are “bell ringer” or ‘warm-up’ games, designed to be played in 5–15 minutes at the beginning or end of class. The other two games take longer, maybe a full class period, but our suggestion is that you play a couple of levels each class until done. The latest game, [Powwow Stands](#), is in answer to the many requests we have received for a game that teaches a lower level of mathematics. This game uses addition to run a couple of powwow stands.



[Follow the bubbles](#) – Designed for grades 4–6. Social Studies content includes discussion and examples primary source, hangman games to teach vocabulary and a couple of math mini-games (tic-tac-toe and caves and ladders) to review multiplication and division. Bonus: This game is 100% accessible for players with visual impairments.



[The Greasy Door](#) teaches the Lakota value of generosity and addresses Oceti Sakowin Essential Understandings Standard 2.3 – Recognize that there is a continuum of tribal identity, ranging from traditional to contemporary lifestyle that includes the challenges of living in two worlds.



[Healing Journeys](#) – Designed for grades 6–8. This short game, tells the story of Dr. Susan La Flesche Picotte, the first Native American doctor, can be played in 15–20 minutes, maybe a little less if your students don't hit too many buffalo with their train.



[Plant Knowledge](#) – Designed for grades 5–7. This short game has four levels teaching about important plants (sage, sweetgrass and prairie turnips), food preservation, Hidatsa agriculture and chokecherries. Teachers from the CIRCLES program [wrote this amazing unit that includes the game.](#)



Thanks to Library of Congress funding, we created 7 games with blocks designed by teachers and now we are moving to having teachers and community members make their own games

<https://www.7generationgames.com/games-teaching-indigenous-rural-history-with-primary-sources/>



[Powwow Stands](#) is our first online game for grades 2–3. Students practice addition while running a powwow stand.

What can you make with blocks?

See examples from our user community



Bake-A-Palooza

Fractions & Fame

Dividing fractions is all that stands between you, baking glory and \$10,000.

[Play](#)

Disaster Deduction Detectives

Learn mean, median and quartiles in context. Data tracking and analysis in real-world scenarios.

[Play](#)

Warrior Vets

Native American history

Using Library of Congress, primary sources learn about veterans from WW I to present.

[Play](#)

Minnesota Turtles

Threatened or Thriving?

More than just math. This environmental science game teaches about local turtle populations.

[Play](#)

So You Think You



Bake-A-Palooza



You Are Here

Math and Science on the



Forgotten Trail

Webinar & mailing list

<https://forms.gle/N2rQ3s9JbiR6muTz9>



<https://www.youtube.com/@7GenBlocks/playlists>



7GenBlocks

@7GenBlocks • 9 subscribers • 19 videos

Want to make an educational game but new to Javascript? Experienced developer who w...more

[7genblocks.com](#) and 1 more link

Subscribed

Videos

Playlists



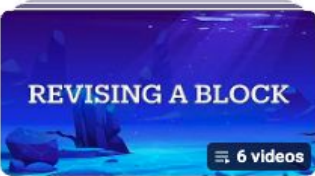
No-code educational game development

View full playlist



Experienced Developer Examples

View full playlist



Getting Started with 7 Gen Blocks

View full playlist

<https://7genblocks.com/>

Make an educational game in one hour

7 Gen Blocks is an educational game development platform that streamlines creating engaging and effective learning games.

No-Code Game Builder

Games Built With Blocks

Build A Game & Win \$1,000





Unblocked

THE NO-CODE GAME COMPETITION

CASH PRIZES

FIRST PLACE

\$500



ZERO CODING
REQUIRED



BUILD AN EDUCATIONAL GAE YOU'D WANT TO PLAY

What can I win?

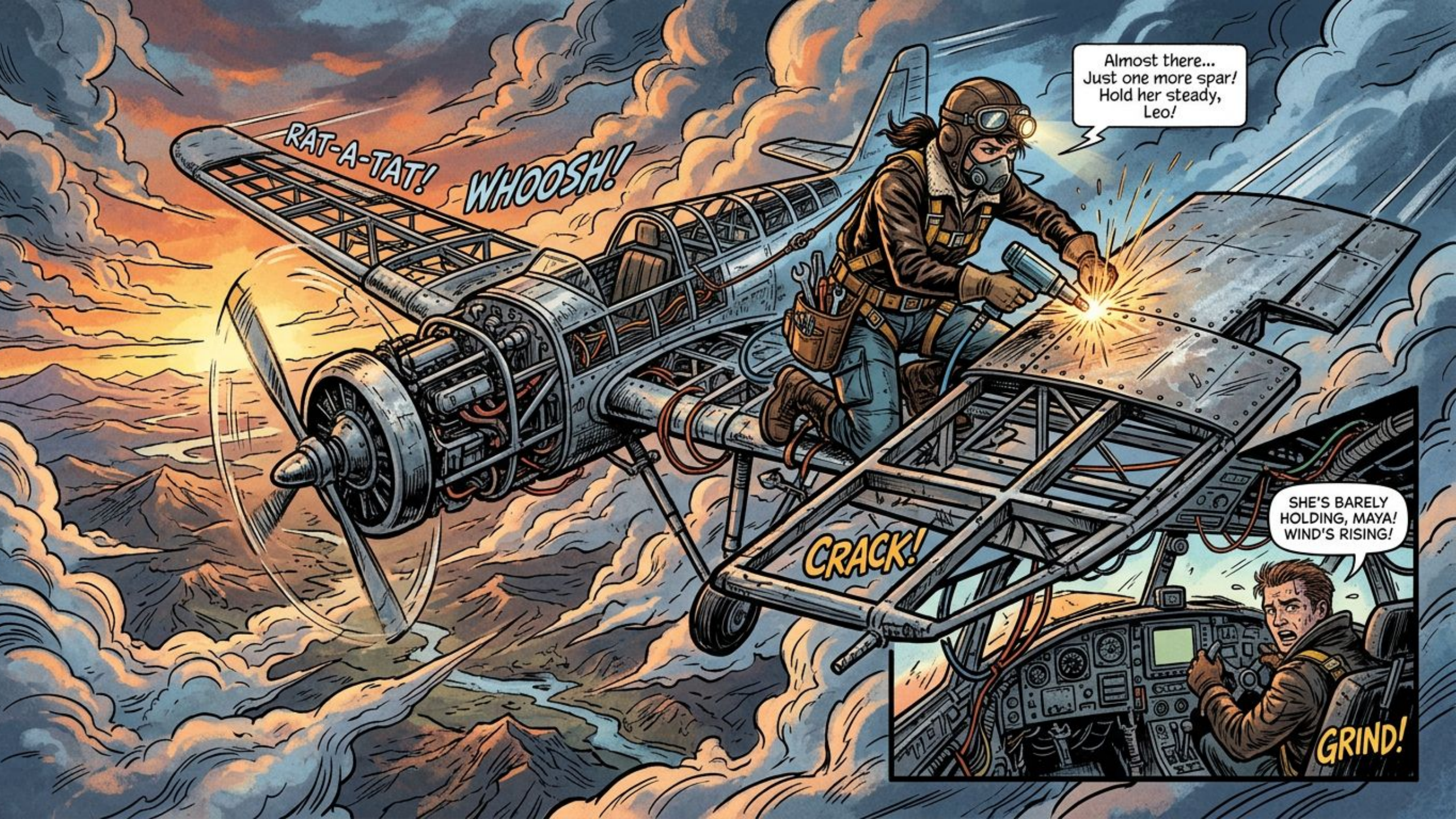
Two prize tracks, and one completed game gets you into both:

Best Game: \$500 first, \$300 second, \$200 third.

Best Artwork: \$500 first, \$300 second, \$200 third.

Didn't win a cash prize but still made a game we like?

Honorable mention: Your choice of 7 Gen Blocks t-shirt, hat or water bottle, plus recognition on our website and the option to have your game published on our Built with Blocks page.



RAT-A-TAT! WHOOSH!

Almost there...
Just one more spar!
Hold her steady,
Leo!

CRACK!

SHE'S BARELY
HOLDING, MAYA!
WIND'S RISING!

GRIND!